

# REALSCOUT

Real Estate Mobile Application

Complete Technical Documentation

A Modern Full-Stack Solution

React Native • Expo • Appwrite • TypeScript

Mausam Kar

Full-Stack Developer & Mobile Enthusiast

Version 1.0.0 | 2024

# Contents

---

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                               | <b>1</b>  |
| 1.1      | Overview . . . . .                                | 1         |
| 1.2      | Key Highlights . . . . .                          | 1         |
| 1.3      | Project Metadata . . . . .                        | 1         |
| <b>2</b> | <b>Features</b>                                   | <b>3</b>  |
| 2.1      | Implemented Features . . . . .                    | 3         |
| <b>3</b> | <b>Architecture</b>                               | <b>4</b>  |
| 3.1      | System Architecture . . . . .                     | 4         |
| 3.2      | Architecture Layers . . . . .                     | 4         |
| <b>4</b> | <b>Technology Stack</b>                           | <b>5</b>  |
| 4.1      | Frontend Technologies . . . . .                   | 5         |
| 4.2      | Backend & Services . . . . .                      | 5         |
| 4.3      | Development Tools . . . . .                       | 5         |
| <b>5</b> | <b>Authentication</b>                             | <b>6</b>  |
| 5.1      | Authentication Flow . . . . .                     | 6         |
| 5.2      | Implementation Code . . . . .                     | 6         |
| <b>6</b> | <b>Database Schema</b>                            | <b>8</b>  |
| 6.1      | Entity Relationship Diagram . . . . .             | 8         |
| 6.2      | Properties Collection . . . . .                   | 8         |
| 6.3      | Additional Collections . . . . .                  | 8         |
| <b>7</b> | <b>Installation &amp; Setup</b>                   | <b>10</b> |
| 7.1      | Prerequisites . . . . .                           | 10        |
| 7.2      | Installation Steps . . . . .                      | 10        |
| 7.2.1    | Step 1: Clone Repository . . . . .                | 10        |
| 7.2.2    | Step 2: Install Dependencies . . . . .            | 10        |
| 7.2.3    | Step 3: Configure Environment Variables . . . . . | 10        |
| 7.2.4    | Step 4: Start Development Server . . . . .        | 11        |
| 7.3      | Appwrite Configuration . . . . .                  | 11        |
| <b>8</b> | <b>Project Structure</b>                          | <b>12</b> |
| 8.1      | Directory Organization . . . . .                  | 12        |
| 8.2      | Component Architecture . . . . .                  | 13        |

|           |                                                         |           |
|-----------|---------------------------------------------------------|-----------|
| <b>9</b>  | <b>Future Roadmap</b>                                   | <b>14</b> |
| 9.1       | Phase 1: Enhanced User Experience (Q1 2025) . . . . .   | 14        |
| 9.2       | Phase 2: Engagement & Notifications (Q2 2025) . . . . . | 14        |
| 9.3       | Phase 3: Advanced Features (Q3 2025) . . . . .          | 15        |
| 9.4       | Phase 4: Social & Integration (Q4 2025) . . . . .       | 15        |
| 9.5       | Long-Term Vision . . . . .                              | 15        |
| <b>10</b> | <b>Application Workflow</b>                             | <b>16</b> |
| 10.1      | User Journey Flowchart . . . . .                        | 16        |
| 10.2      | Data Flow Pipeline . . . . .                            | 16        |
| <b>11</b> | <b>Advantages &amp; Benefits</b>                        | <b>18</b> |
| 11.1      | Technical Advantages . . . . .                          | 18        |
| 11.2      | Business Advantages . . . . .                           | 18        |
| 11.3      | User Experience Benefits . . . . .                      | 19        |
| <b>12</b> | <b>Conclusion</b>                                       | <b>20</b> |
| 12.1      | Summary . . . . .                                       | 20        |
| 12.2      | Key Takeaways . . . . .                                 | 20        |
| 12.3      | Future Outlook . . . . .                                | 20        |
| <b>A</b>  | <b>Additional Resources</b>                             | <b>21</b> |
| A.1       | Useful Links . . . . .                                  | 21        |
| A.2       | License Information . . . . .                           | 21        |

# 1

# Introduction

---

## 1.1 Overview

---

**Real Scout** is a modern, full-stack real estate mobile application built with React Native and Expo. It provides users with a seamless experience to browse, search, and explore properties with an intuitive interface and powerful features.

### Key Point

The application leverages **Google OAuth** for secure authentication and **Appwrite** as a backend-as-a-service platform for managing data, authentication, and storage.

## 1.2 Key Highlights

---

- 🔒 **Secure Authentication** with Google OAuth 2.0
- 🏠 **Dynamic Property Listings** with advanced search and filtering
- 📱 **Cross-platform** support (iOS & Android)
- 🎮 **Modern UI/UX** with NativeWind (Tailwind CSS for React Native)
- ⚡ **Fast Performance** with optimized data fetching
- 🔄 **Real-time Updates** with Appwrite
- 🌱 **Database Seeding** utility for quick setup

## 1.3 Project Metadata

---

| Attribute  | Details                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------|
| Version    | 1.0.0                                                                                                     |
| License    | MIT License                                                                                               |
| Platform   | iOS & Android                                                                                             |
| Repository | <a href="https://github.com/Mausam5055/Real-Estate-APP">https://github.com/Mausam5055/Real-Estate-APP</a> |
| Author     | Mausam Kar                                                                                                |
| Status     | Production Ready                                                                                          |

Table 1.1: Project Metadata

# 2 Features

## 2.1 Implemented Features

The following table presents a comprehensive overview of all implemented and planned features:

| Feature               | Description                                         | Status |
|-----------------------|-----------------------------------------------------|--------|
| Google Authentication | Seamless OAuth 2.0 login with Google                | ✓      |
| Property Browsing     | Browse all available properties with pagination     | ✓      |
| Advanced Search       | Search properties by name, address, or type         | ✓      |
| Smart Filters         | Filter properties by type (House, Apartment, Villa) | ✓      |
| Property Details      | Comprehensive property info with image gallery      | ✓      |
| User Profiles         | Manage user settings and preferences                | ✓      |
| Agent Information     | View detailed agent profiles and contact info       | ✓      |
| Reviews & Ratings     | Read property reviews and ratings                   | ✓      |
| Facilities Overview   | View property amenities (Gym, Pool, Parking)        | ✓      |
| Database Seeding      | Quick setup with sample data                        | ✓      |
| Responsive Design     | Optimized for all screen sizes                      | ✓      |
| Offline Support       | Cache data for offline viewing                      | ⚙️     |
| Push Notifications    | Get notified about new properties                   | 🕒      |
| Favorites/Wishlist    | Save properties for later viewing                   | 🕒      |
| Property Comparison   | Compare multiple properties side-by-side            | 🕒      |

Table 2.1: Feature Implementation Status

Legend: ✓ Implemented | ⚙️ In Progress | 🕒 Planned

# 3

## Architecture

### 3.1 System Architecture

Real Scout follows a clean, scalable architecture pattern with clear separation of concerns across multiple layers.

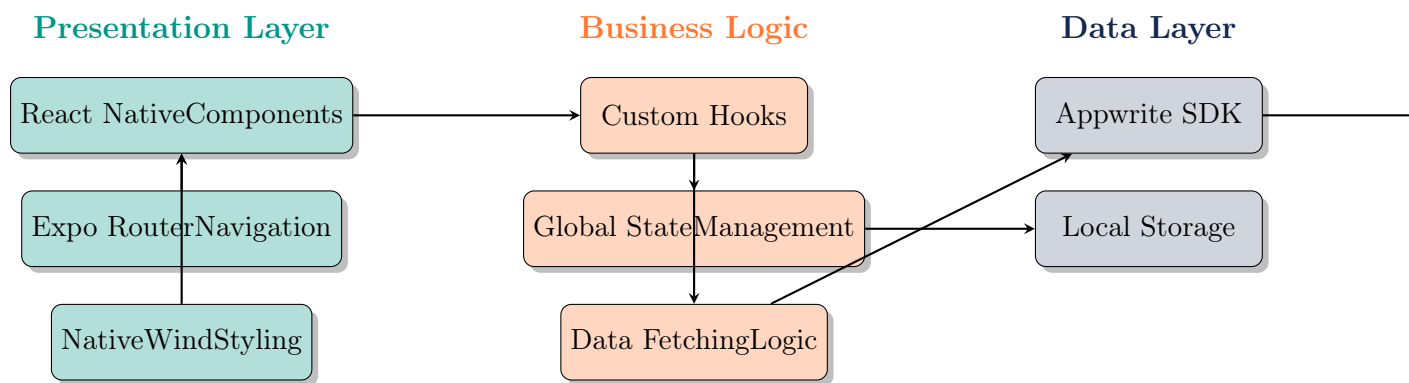


Figure 3.1: System Architecture Diagram

### 3.2 Architecture Layers

1. **Presentation Layer:** React Native components styled with NativeWind providing the user interface
2. **Business Logic Layer:** Custom hooks and state management handling application logic
3. **Data Layer:** Appwrite SDK integration with local caching for offline support
4. **Backend Services:** Appwrite cloud services for authentication, database, and storage

# 4 Technology Stack

## 4.1 Frontend Technologies

| Technology       | Version | Purpose                            |
|------------------|---------|------------------------------------|
| React Native     | 0.81.5  | Cross-platform mobile framework    |
| Expo             | ~54.0.0 | Development platform and toolchain |
| TypeScript       | 5.3.3   | Type-safe JavaScript               |
| NativeWind       | 4.1.23  | Tailwind CSS for React Native      |
| Expo Router      | ~6.0.19 | File-based routing                 |
| React Navigation | 7.0.0   | Navigation library                 |

Table 4.1: Frontend Technology Stack

## 4.2 Backend & Services

| Technology         | Purpose                                  |
|--------------------|------------------------------------------|
| Appwrite           | Backend-as-a-Service (BaaS) platform     |
| Google OAuth       | Authentication provider for secure login |
| Appwrite Databases | NoSQL document database                  |
| Appwrite Storage   | File storage and CDN                     |

Table 4.2: Backend Technology Stack

## 4.3 Development Tools

| Tool   | Purpose                  |
|--------|--------------------------|
| Jest   | Unit testing framework   |
| ESLint | Code linting and quality |
| Babel  | JavaScript compiler      |
| Metro  | JavaScript bundler       |

Table 4.3: Development Tools



# 5

## Authentication

### 5.1 Authentication Flow

The application uses Google OAuth 2.0 for secure user authentication. The following sequence diagram illustrates the complete authentication process:

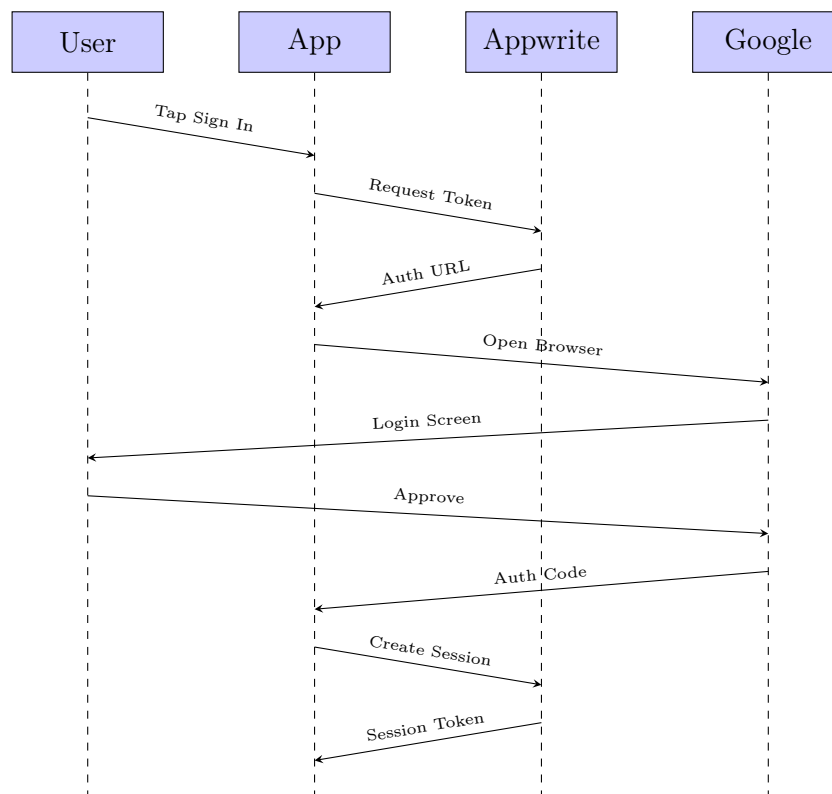


Figure 5.1: OAuth 2.0 Authentication Sequence

### 5.2 Implementation Code

```
1 // lib/appwrite.ts
2 export async function login() {
3   const redirectUri = Linking.createURL("/");
4   const response = await account.createOAuth2Token(
5     OAuthProvider.Google,
6     redirectUri
```

```
7   );
8
9   const browserResult = await openAuthSessionAsync(
10     response.toString(),
11     redirectUri
12   );
13
14   const url = new URL(browserResult.url);
15   const secret = url.searchParams.get("secret");
16   const userId = url.searchParams.get("userId");
17
18   const session = await account.createSession(userId, secret);
19   return session;
20 }
```

Listing 5.1: Authentication Implementation

# 6

## Database Schema

---

### 6.1 Entity Relationship Diagram

---

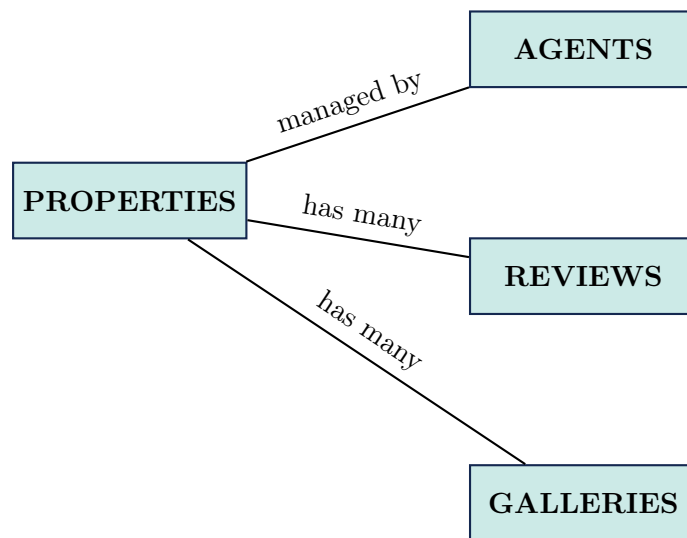


Figure 6.1: Database Entity Relationship Diagram

### 6.2 Properties Collection

---

### 6.3 Additional Collections

---

#### Definition

**Agents Collection:** Stores agent information including name, email, and avatar.

**Reviews Collection:** Contains user reviews with name, avatar, review text, and rating.

**Galleries Collection:** Stores additional property images for gallery view.

| Field       | Type         | Description                        | Required |
|-------------|--------------|------------------------------------|----------|
| name        | String       | Property name                      | Yes      |
| type        | String       | Property type (House, Villa, etc.) | Yes      |
| description | Text         | Detailed description               | Yes      |
| address     | String       | Physical address                   | Yes      |
| geolocation | String       | Lat/Long coordinates               | No       |
| price       | Number       | Property price                     | Yes      |
| area        | Number       | Area in square feet                | Yes      |
| bedrooms    | Number       | Number of bedrooms                 | Yes      |
| bathrooms   | Number       | Number of bathrooms                | Yes      |
| rating      | Number       | Average rating (1-5)               | Yes      |
| facilities  | Array        | List of amenities                  | No       |
| image       | String       | Main property image URL            | Yes      |
| agent       | Relationship | Related agent ID                   | Yes      |
| reviews     | Relationship | Array of review IDs                | No       |
| gallery     | Relationship | Array of gallery IDs               | No       |

Table 6.1: Properties Collection Schema

# 7 Installation & Setup

---

## 7.1 Prerequisites

---

Before starting, ensure you have the following installed:

- **Node.js** (v18 or higher)
- **npm** or **yarn** package manager
- **Git** version control
- **Expo Go** app on your mobile device

## 7.2 Installation Steps

---

### 7.2.1 Step 1: Clone Repository

```
1 git clone https://github.com/Mausam5055/Real-Estate-APP.git
2 cd Real-Estate-APP
```

### 7.2.2 Step 2: Install Dependencies

```
1 npm install
2 # or
3 yarn install
```

### 7.2.3 Step 3: Configure Environment Variables

Create a `.env.local` file in the root directory:

```
1 EXPO_PUBLIC_APPWRITE_ENDPOINT=https://cloud.appwrite.io/v1
2 EXPO_PUBLIC_APPWRITE_PROJECT_ID=your_project_id
3 EXPO_PUBLIC_APPWRITE_DATABASE_ID=your_database_id
4 EXPO_PUBLIC_APPWRITE_GALLERIES_COLLECTION_ID=your_galleries_id
5 EXPO_PUBLIC_APPWRITE_REVIEWS_COLLECTION_ID=your_reviews_id
6 EXPO_PUBLIC_APPWRITE_AGENTS_COLLECTION_ID=your_agents_id
7 EXPO_PUBLIC_APPWRITE_PROPERTIES_COLLECTION_ID=your_properties_id
8 EXPO_PUBLIC_APPWRITE_BUCKET_ID=your_bucket_id
```

### 7.2.4 Step 4: Start Development Server

```
1 npm start
2 # or
3 npx expo start
```

## 7.3 Appwrite Configuration

1. Create an account at <https://cloud.appwrite.io/>
2. Create a new project
3. Set up collections (Properties, Agents, Reviews, Galleries)
4. Enable Google OAuth provider
5. Copy credentials to `.env.local` file

### Note

**Database Seeding:** After setup, use the built-in seeding utility to populate your database with sample data. Access this feature from the Profile section in the app.

## 8

## Project Structure

## 8.1 Directory Organization

```

1 Real-Estate-APP/
2   app/                                # Application screens
3     (root)/                          # Protected routes
4       (tabs)/                        # Bottom tab navigation
5         index.tsx                    # Home screen
6         explore.tsx                  # Explore screen
7         profile.tsx                  # Profile screen
8         properties/                  # Property details
9           [id].tsx                    # Dynamic detail page
10      _layout.tsx                     # Root layout
11      sign-in.tsx                     # Auth screen
12      seed-data.tsx                   # DB seeding utility
13  assets/                             # Static assets
14    fonts/                            # Custom fonts
15    icons/                            # Icon images
16    images/                           # App images
17  components/                         # Reusable components
18    Cards.tsx                         # Property cards
19    Comment.tsx                       # Review comments
20    Filters.tsx                       # Filter component
21    NoResults.tsx                     # Empty state
22    Search.tsx                        # Search bar
23  constants/                          # App constants
24    icons.ts                          # Icon exports
25    images.ts                         # Image exports
26    data.ts                           # Static data
27  lib/                                # Core utilities
28    appwrite.ts                       # Appwrite config
29    global-provider.tsx               # Global state
30    useAppwrite.ts                    # Data hook
31    seed.ts                           # Seeding logic
32    data.ts                           # Sample data
33  .env.local                          # Environment vars
34  app.json                            # Expo config
35  package.json                        # Dependencies
36  tailwind.config.js                  # Tailwind config
37  tsconfig.json                       # TypeScript config

```

Listing 8.1: Project Directory Structure

## 8.2 Component Architecture

---

### Key Point

The application follows a **modular component architecture** where each component is responsible for a single functionality. This promotes code reusability and maintainability.



# 9

## Future Roadmap

---

### 9.1 Phase 1: Enhanced User Experience (Q1 2025)

---

- **Favorites/Wishlist Feature**
  - Save properties for later viewing
  - Sync across devices
  - Share wishlist with others
- **Advanced Filters**
  - Price range slider
  - Multiple facility selection
  - Sort by distance, price, rating
- **Property Comparison**
  - Compare up to 3 properties side-by-side
  - Visual comparison charts
  - Export comparison as PDF

### 9.2 Phase 2: Engagement & Notifications (Q2 2025)

---

- **Push Notifications**
  - New property alerts based on preferences
  - Price drop notifications
  - Saved search alerts
- **In-App Messaging**
  - Direct chat with agents
  - Schedule property viewings
  - Real-time message notifications
- **Virtual Tours**
  - 360° property views
  - Video walkthroughs
  - AR room visualization

### 9.3 Phase 3: Advanced Features (Q3 2025)

---

- **AI-Powered Recommendations**
  - Personalized property suggestions
  - Smart search based on user behavior
  - Price prediction models
- **Mortgage Calculator**
  - EMI calculator
  - Loan eligibility checker
  - Bank comparison
- **Offline Mode**
  - Cache property data for offline viewing
  - Sync when back online
  - Download property brochures

### 9.4 Phase 4: Social & Integration (Q4 2025)

---

- Social media sharing
- Community reviews and ratings
- Google Maps integration
- Calendar integration for appointments
- Analytics dashboard

### 9.5 Long-Term Vision

---

| Feature                | Description                                |
|------------------------|--------------------------------------------|
| Multi-Language Support | Add support for regional languages         |
| Dark Mode              | Complete dark theme implementation         |
| Web Platform           | Expand to web using React.js               |
| Admin Dashboard        | Property management portal for agents      |
| Payment Gateway        | Integrate booking/token payment            |
| Blockchain Integration | Property ownership verification using NFTs |

Table 9.1: Long-Term Vision

# 10 Application Workflow

---

## 10.1 User Journey Flowchart

---

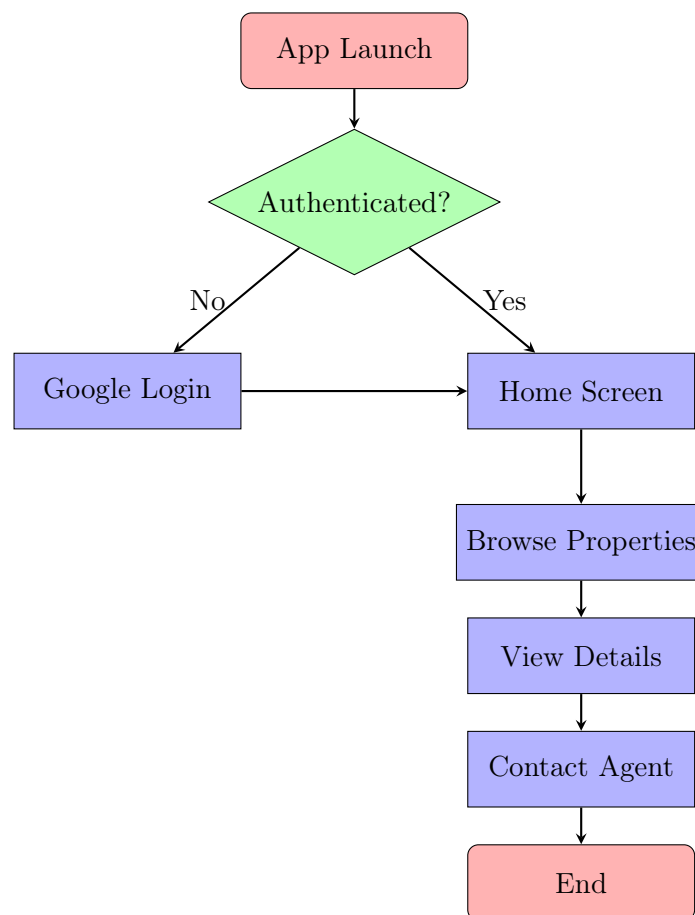


Figure 10.1: User Journey Flowchart

## 10.2 Data Flow Pipeline

---



Figure 10.2: Data Flow Pipeline

# 11 Advantages & Benefits

---

## 11.1 Technical Advantages

---

### 1. Cross-Platform Development

- Single codebase for iOS and Android
- Reduced development time and cost
- Consistent user experience across platforms

### 2. Type Safety with TypeScript

- Catch errors during development
- Better IDE support and autocomplete
- Improved code maintainability

### 3. Backend-as-a-Service

- No backend infrastructure management
- Built-in authentication and database
- Automatic scaling and security

### 4. Modern UI Framework

- Rapid UI development with NativeWind
- Consistent design system
- Responsive layouts

## 11.2 Business Advantages

---

### Key Point

Real Scout provides significant business value through:

- Faster time-to-market
- Lower development costs
- Scalable architecture
- Easy maintenance and updates
- Rich feature set out of the box

### 11.3 User Experience Benefits

| Benefit          | Impact                             |
|------------------|------------------------------------|
| Fast Performance | Quick property browsing and search |
| Intuitive UI     | Easy navigation for all user types |
| Secure Login     | Peace of mind with Google OAuth    |
| Rich Information | Comprehensive property details     |
| Visual Gallery   | High-quality property images       |
| Agent Profiles   | Direct communication with agents   |

Table 11.1: User Experience Benefits

# 12 Conclusion

---

## 12.1 Summary

---

Real Scout represents a modern approach to real estate mobile applications, combining cutting-edge technologies with user-centric design. The application successfully demonstrates:

- Effective use of React Native for cross-platform development
- Seamless integration with Appwrite backend services
- Secure authentication through Google OAuth
- Scalable and maintainable code architecture
- Rich feature set for property browsing and exploration

## 12.2 Key Takeaways

---

### Definition

**React Native + Appwrite:** This combination provides a powerful stack for rapid mobile app development with minimal backend management.

**TypeScript:** Adds type safety and improves developer experience significantly.

**NativeWind:** Brings the productivity of Tailwind CSS to React Native development.

## 12.3 Future Outlook

---

The roadmap for Real Scout includes exciting features such as AI-powered recommendations, virtual tours, and blockchain integration. These enhancements will further solidify its position as a comprehensive real estate solution.

### Note

This project serves as an excellent foundation for anyone looking to build production-ready mobile applications using modern technologies and best practices.



# Additional Resources

---

## A.1 Useful Links

---

- GitHub Repository: <https://github.com/Mausam5055/Real-Estate-APP>
- React Native Documentation: <https://reactnative.dev/>
- Expo Documentation: <https://docs.expo.dev/>
- Appwrite Documentation: <https://appwrite.io/docs>
- NativeWind Documentation: <https://www.nativewind.dev/>

## A.2 License Information

---

This project is licensed under the MIT License. Copyright (c) 2024 Mausam Kar.

**Thank you for reading!**

*For questions or contributions, please visit the GitHub repository.*